BOSTON HOUSING PRICE PREDICTION USING LINEAR REGRESSION

Course:	Machine Learning / Data Science
Project:	Boston Housing Regression Model
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Date:	Monday 9th March 2026

1. INTRODUCTION

Background of the Problem

Housing price prediction is an important problem in the real estate industry. Property prices depend on several factors such as the number of rooms, crime rate, property taxes, and access to services. Understanding these factors helps property buyers, investors, and real estate agents make informed decisions.

However, manually analyzing these variables can be difficult because housing prices are influenced by many factors simultaneously. Machine learning techniques provide efficient methods for analyzing large datasets and identifying patterns that influence property prices.

Importance of Machine Learning

Machine learning enables computers to learn patterns from historical data and make predictions. In the real estate sector, machine learning models can analyze housing characteristics and estimate property prices automatically.

This approach improves decision-making, reduces human effort, and provides more accurate insights into housing markets.

Objective of the Project

The objective of this project is to build a Linear Regression model to predict housing prices using the Boston Housing dataset. The project also involves performing exploratory data analysis, training a predictive model, evaluating model performance, and deploying the model using a Streamlit web application.

2. DATASET DESCRIPTION

Dataset Source

The dataset used in this project is the Boston Housing dataset, which contains housing information for suburbs in Boston. The dataset includes several variables that describe housing conditions and neighborhood characteristics.

Description of Variables

Variable	Description
CRIM	Crime rate per town
ZN	Residential land proportion
INDUS	Non-retail business acres
CHAS	Charles River proximity
NOX	Nitric oxide concentration
RM	Average number of rooms
AGE	Houses built before 1940
DIS	Distance to employment centers
RAD	Highway accessibility
TAX	Property tax rate
PTRATIO	Student-teacher ratio
B	Demographic indicator
LSTAT	Lower status population (%)

Target Variable

The target variable is PRICE, representing the median value of owner-occupied homes.

Justification of Target Variable

MEDV is a continuous numerical variable, which makes it suitable for regression analysis. The goal of this project is to predict housing prices based on housing features, making Linear Regression an appropriate modeling technique.

3. EXPLORATORY DATA ANALYSIS (EDA)

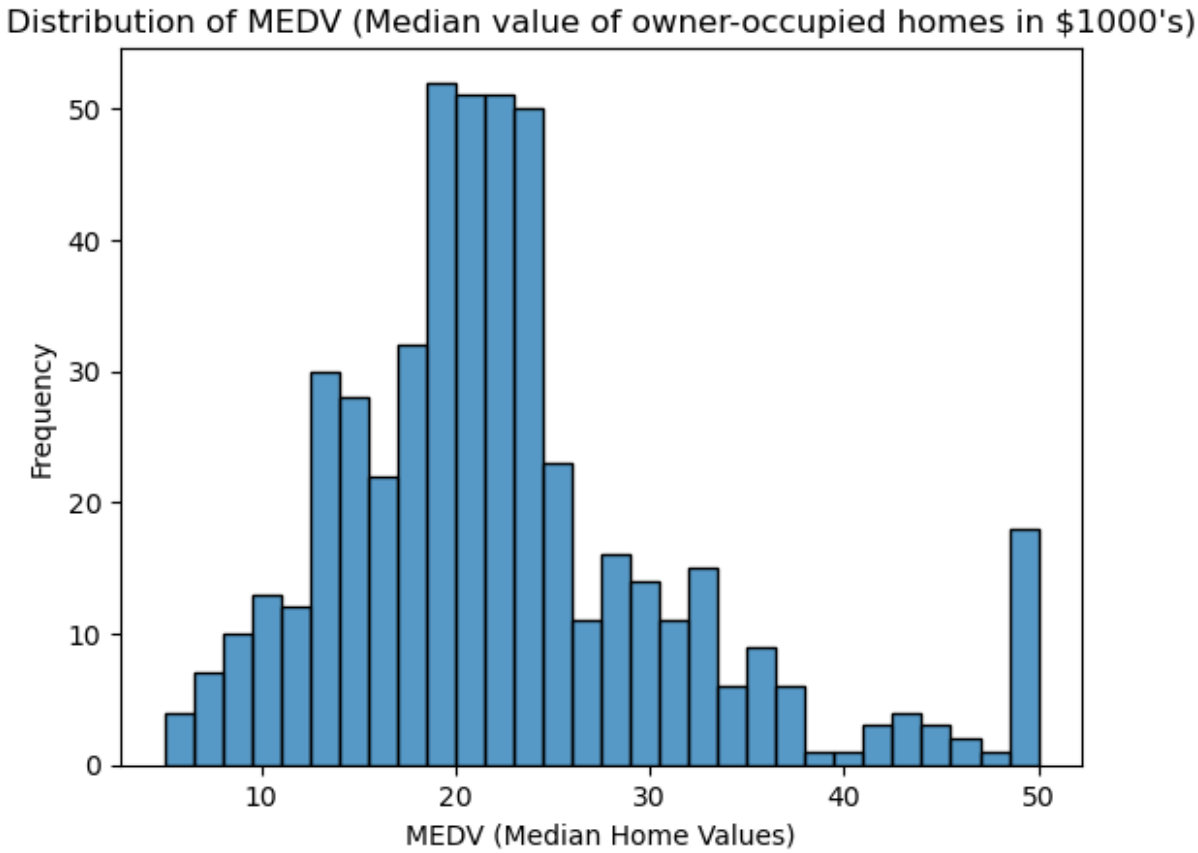
EDA was performed to understand the structure, distribution, and relationships within the dataset.

Statistical Analysis

Basic statistical measures such as mean, median, standard deviation, and correlation were calculated to understand the dataset. These statistics helped identify patterns and relationships between variables.

Key Visualizations:

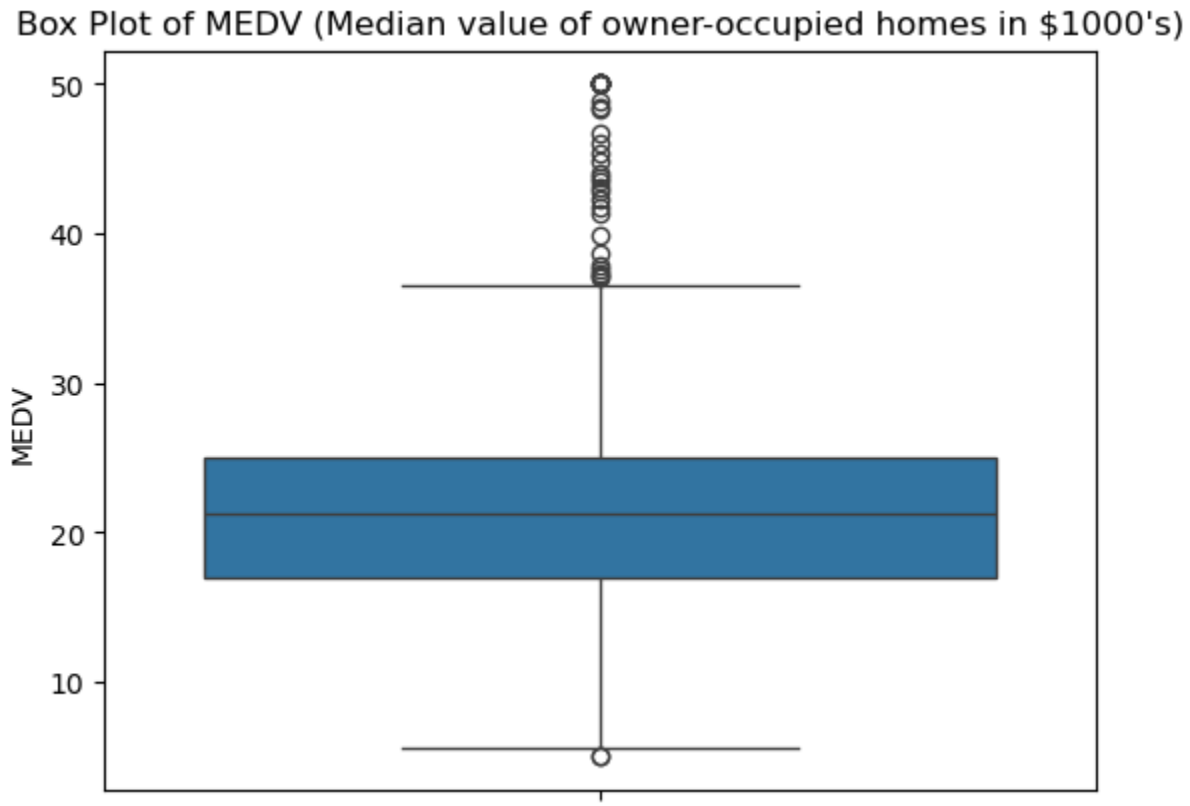
1.Histogram of House Prices



Interpretation;

The histogram shows the distribution of house prices. Most houses fall within moderate price ranges, while fewer houses have extremely high prices.

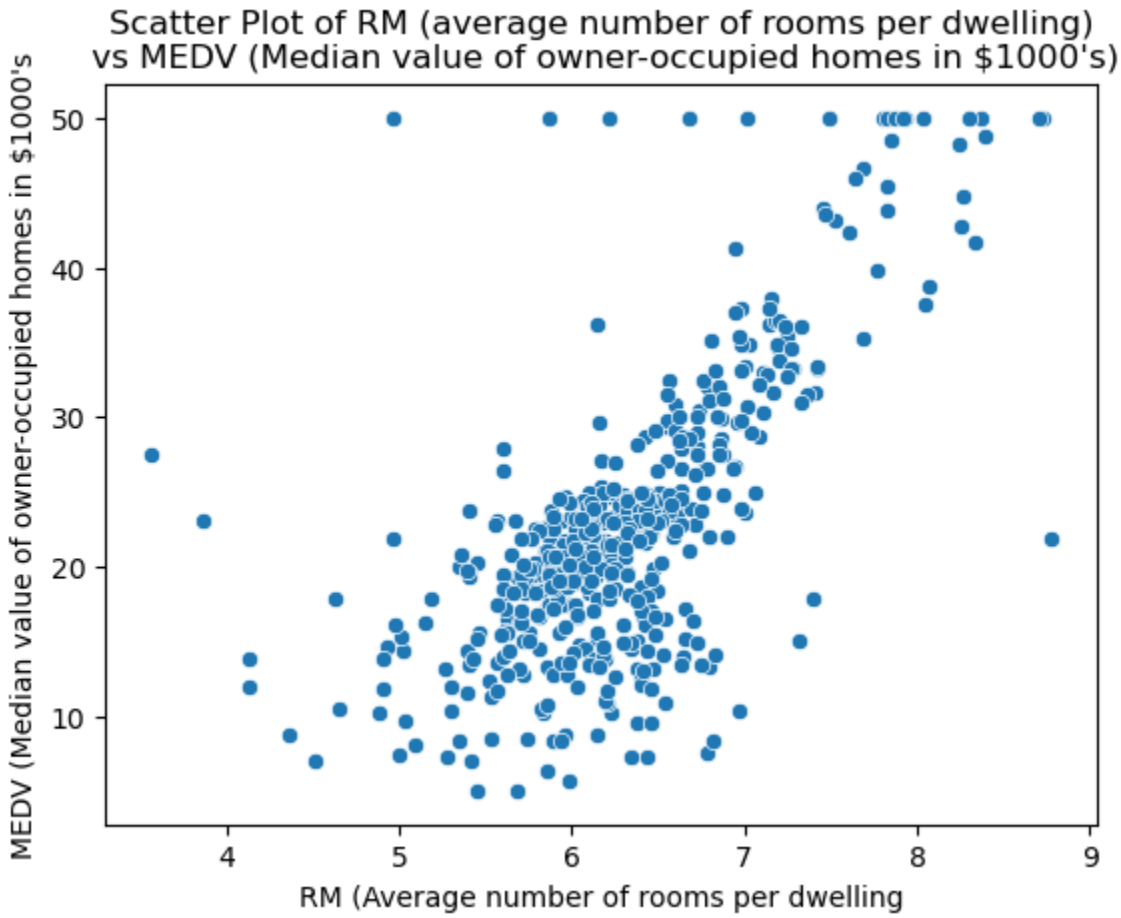
2. Box Plot of MEDV



Interpretation:

The box plot shows the spread of housing prices and highlights possible outliers.

3. Scatter Plot (RM vs MEDV)



Interpretation:

The scatter plot shows a positive relationship between the number of rooms and house price. This means that houses which have more rooms tend to have high prices than houses with lesser rooms.

4. Correlation Heatmap

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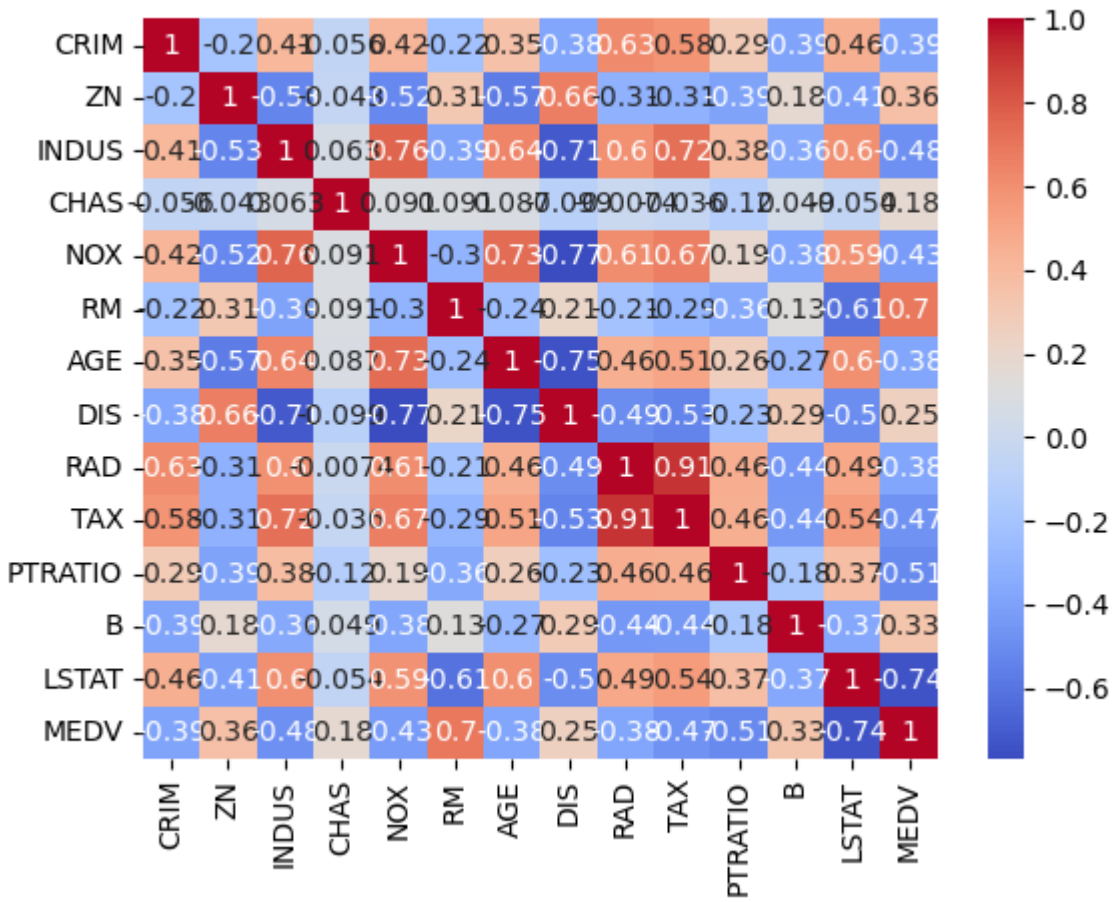

Interpretation:

The heat map shows correlations between variables. RM shows strong positive correlation with MEDV, while LSTAT shows strong negative correlation with MEDV.

Key observation

The exploratory analysis revealed several important insights:

- The variable RM (average number of rooms**)** has a strong positive relationship with housing prices.

- The variable LSTAT (percentage of lower-status population) has a strong negative relationship with housing prices.
- Some variables show significant variation and contain potential outliers.
- Most housing prices are concentrated within a moderate range rather than extreme values.

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4. MODEL DEVELOPMENT

Data Preprocessing

Several preprocessing steps were performed before training the model:

- Feature selection
- Train-test split (80% training, 20% testing)
- Feature scaling
- Outlier detection and treatment

These steps ensure that the dataset is suitable for machine learning.

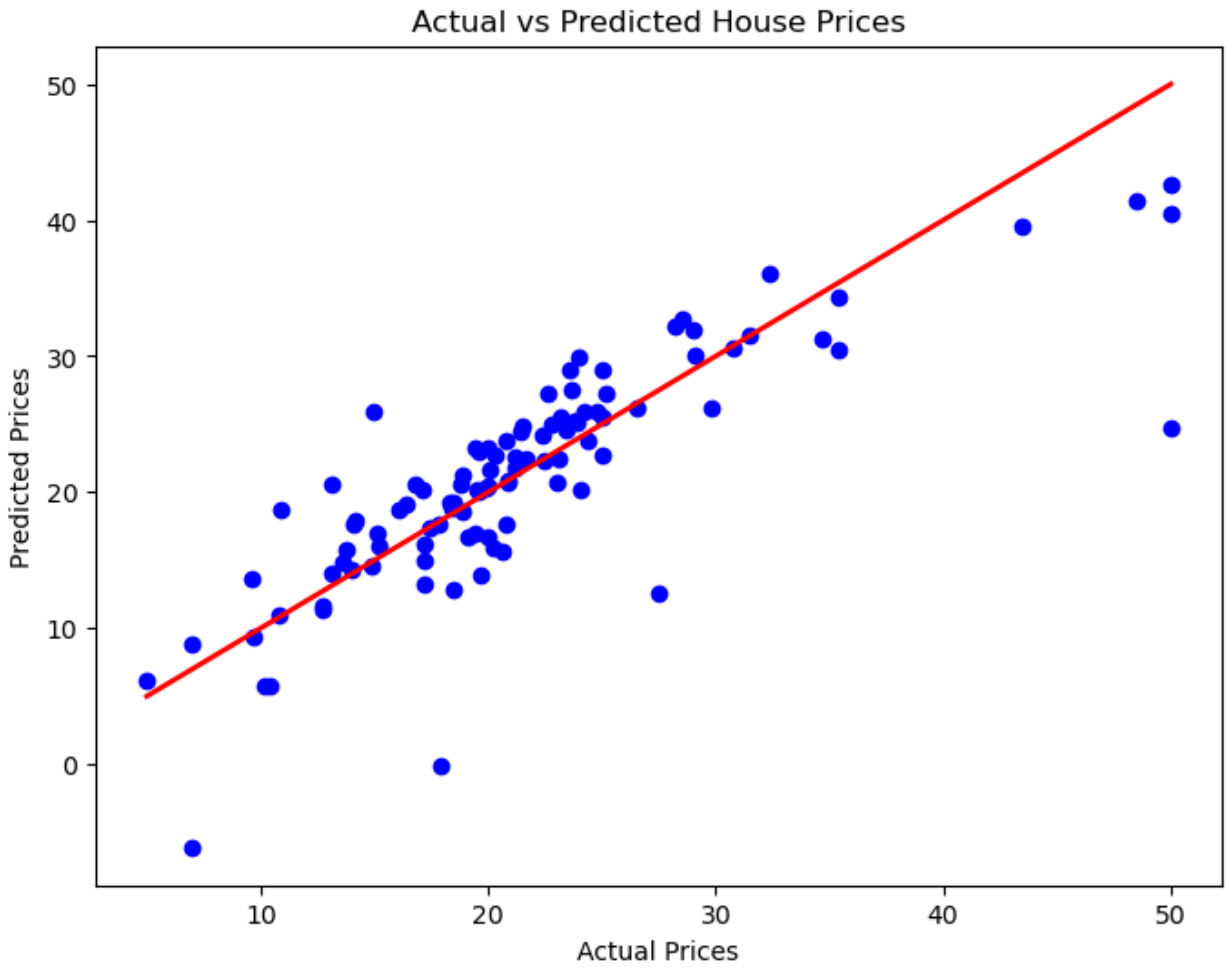
Model Training

A Linear Regression model was used to train the dataset. The model learns the relationship between housing features and house prices.

Model Evaluation

The following evaluation metrics were used:

Metric	Description
MAE	Average prediction error
MSE	Squared prediction error
RMSE	Error in same unit as price
R^2 Score	Model accuracy level



The scatter plot compares the actual house prices with the predicted prices generated by the Linear Regression model. Points that lie closer to the diagonal line indicate accurate predictions, while points farther from the line represent prediction errors. The distribution of points around the diagonal line suggests that the model captures the general relationship between housing features and house prices.

5. FINDINGS AND INSIGHTS

Several important insights were discovered from the analysis:

Insights from Data Analysis

The dataset shows that housing prices are influenced by multiple factors including number of rooms, crime rate, and tax rates.

Insights from Visualizations

Visualizations revealed strong relationships between certain variables and house prices. In particular:

- Houses with more rooms tend to have higher prices.
- Areas with higher percentages of lower-status population tend to have lower housing prices.

Insights from Model Performance

The Linear Regression model was able to capture the relationship between housing features and house prices. The evaluation metrics indicate that the model performs reasonably well in predicting house prices based on the available data.

6. RECOMMENDATIONS

Based on the findings of this project, the following recommendations can be made:

- Additional features such as location, infrastructure, and economic conditions could improve prediction accuracy.
- More advanced machine learning models such as Random Forest or Gradient Boosting could be explored for better performance.
- Larger and more recent housing datasets could be used to improve prediction reliability.
- The deployed Streamlit application could be enhanced with more interactive features for users.

7. CONCLUSION

This project demonstrated how machine learning can be used to predict housing prices using the Boston Housing dataset. The project involved data exploration, visualization, preprocessing, model training, and model deployment using a Streamlit web application.

The results show that Linear Regression can effectively model the relationship between housing characteristics and house prices. Variables such as the number of rooms and population characteristics were found to significantly influence housing prices.

Future improvements could include using more advanced algorithms and incorporating additional real-world features to improve prediction accuracy.